Diversity-Coverage Trade-Offs in Cold-Start Recommendations*

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Abstract

The cold-start problem is a fundamental bottleneck in recommender systems, where the vast majority of users and items have little to no interaction history. Under these extreme conditions, standard collaborative filtering models collapse, while Graph Neural Networks (GNNs) suffer from message-passing breakdown over fragmented topologies. To overcome these challenges, we present an end-to-end framework that addresses cold-start constraints through two mechanisms: a feature-guided graph enrichment, which densifies user-item graphs using supervised propensity modeling and Explainable AI (XAI) feature attribution, and a scalable re-ranking strategy designed to enhance recommendation diversity. Evaluated on a production telecommunications catalog encompassing over one million users, our framework doubles Intra-List Diversity (ILD) while maintaining over 95% exposure for cold items.

Keywords

Recommender Systems, Extreme Cold-Start, Graph Neural Networks, Diversity Re-ranking

1. Introduction

Modern industrial recommenders struggle when deployed across sparse environments where a large fraction of consumers and catalog items lack recorded interactions [2, 3]. While collaborative filtering (CF) algorithms demonstrate strong performance in dense interaction regimes, they degrade severely when confronted with isolated nodes. Although Graph Neural Networks (GNNs) [4, 5, 6] improve multi-hop collaborative signal discovery, their propagation mechanics fail when graphs lack structural edges, preventing cold users from obtaining meaningful latent embeddings.

Addressing extreme cold-start settings exposes an operational trade-off between relevance, catalog coverage, and diversity. Standard optimization criteria inadvertently amplify popular items (the Matthew effect [7]) and trap users to repetitive interaction patterns (filter bubbles [8]). Conversely, forcing exposure for long-tail catalog items with minimal interaction evidence frequently degrades top-K ranking precision. Crucially, we observe a direct geometric connection between diversity and coverage: enforcing diversity requires exploring distant regions of the latent space, which inherently expands exposure across the wider catalog. The central challenge lies in navigating this space without sacrificing recommendation quality.

To address this challenge, we introduce an end-to-end framework tailored for large-scale enterprise environments. The pipeline transforms disconnected interaction graphs into densely populated topologies via a XAI-guided edge augmentation module, subsequently trains an inductive GNN to capture structural representations, and finally applies a centroid-aware re-ranking strategy to maximize recommendation diversity. We evaluate our methodology using a proprietary dataset from TIM S.p.A., a major telecommunications provider, consisting of over one million customers where 83% exhibit zero transaction history.

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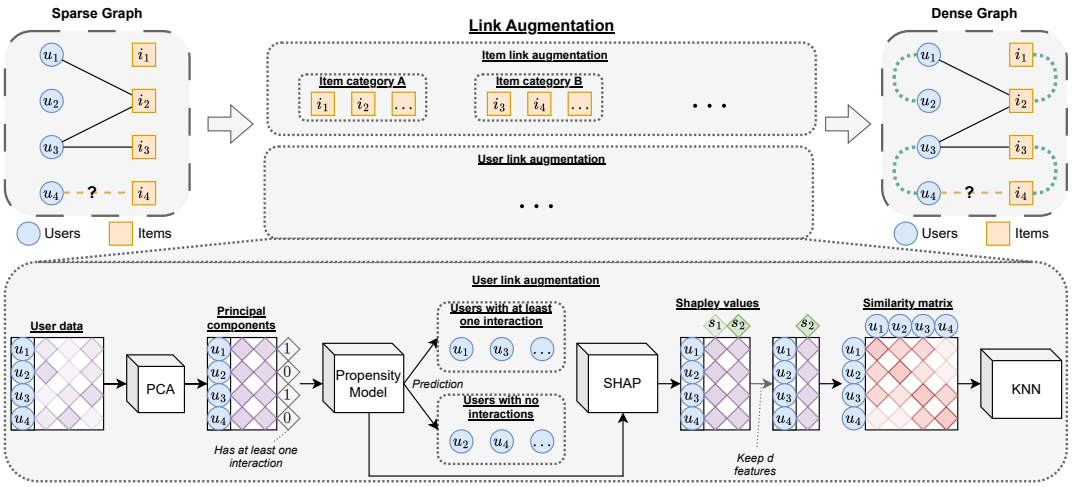


Figure 1: The graph augmentation module leverages a purchase propensity model to make user feature selection and densify the binary graph through KNN.

2. Methodology

Our framework is composed by three main modules: (i) graph augmentation, (ii) inductive graph representation learning, and (iii) a centroid-aware diversity re-ranker.

2.1. Graph Augmentation

Let $G = (U, I, E)$ be a bipartite graph where $U = \{u_1, \ldots, u_N\}$ represents users, $I = \{i_1, \ldots, i_M\}$ denotes items, and $E \subseteq U \times I$ corresponds to historical purchases. In cold-start regimes, $|E| \ll |U| \times |I|$, resulting in a high prevalence of isolated nodes.

To restore message propagation while preserving graph integrity, we augment homogeneous edges. Each user is initially described by high-dimensional vectors $\mathbf{z}_u \in \mathbb{R}^D$ ($D > 600$). To construct the connections efficiently, we first compress the high-dimensional feature vectors $\mathbf{z}_u \in \mathbb{R}^D$ via Principal Component Analysis (PCA), yielding lower-dimensional representations $\mathbf{x}_u \in \mathbb{R}^m$ ($m \ll D$) that preserve primary behavioral variance while mitigating noise. Next, an XGBoost model predicts interaction propensity $\hat{y}_u = \sigma(\sum_t f_t(\mathbf{x}_u))$ under majority-class undersampling. Rather than evaluating similarities over an unweighted or noisy feature space, we leverage Explainable AI (XAI) to interpret this model. Specifically, we compute Shapley values $\phi_j(u)$ across predictions and extract the top-d attributes that maximize aggregate attribution: $F_{\text{top}} = \arg\max_{S,|S|=d} \sum_{j \in S} \sum_u |\phi_j(u)|$. Finally, we index users over this refined subspace using Annoy random projection trees, connecting each node to its top-k nearest counterpart to augment user-user edges E_{UU}. Symmetrically, item-item links E_{II} are built by connecting items that share common catalog features. The resulting dense graph $\tilde{G} = (U \cup I, E \cup E_{UU} \cup E_{II})$ allows inductive propagation across all catalog entities.

2.2. Inductive Structural Representation

We deploy an inductive GraphSAGE [5] backbone over the augmented graph. Choosing an inductive architecture over transductive approaches (e.g., LightGCN) is critical for industrial viability: once our augmentation step connects an inactive customer to the graph via synthetic edges E_{UU}, GraphSAGE immediately computes their embedding $\mathbf{h}_u \in \mathbb{R}^{d_e}$ by aggregating the representations of connected peers, bypassing expensive model retraining. We train the model via mini-batch link prediction, scoring candidate pairs via dot product $\hat{s}_{ui} = \mathbf{h}_u^\top \mathbf{h}_i$ and optimizing a class-weighted cross-entropy loss with dynamic negative sampling to counteract the severe purchase imbalance. By propagating collaborative signals across both original interactions and augmented links, the network projects cold users and unsold items into a shared latent space, providing the initial recommendations for our re-ranking step.

2.3. Centroid-Aware Diversity Re-ranking

Link prediction over the learned GNN representations inherently reinforces popularity bias, disproportionately favoring frequently purchased items at the expense of recommendation diversity. Mitigating this exposure skew typically requires re-ranking; however, evaluating and diversifying candidate sets individually across millions of customers introduces prohibitive computational latency. To overcome this operational bottleneck, we introduce a cluster-centroid diversification mechanism.

First, we partition the user latent space $\{\mathbf{h}_u\}_{u \in U}$ into K_c clusters using k-means||, determining cluster count via Silhouette optimization. For each cluster C_k, the user closest to centroid $\mathbf{q}_k$ acts as cluster representative, generating centroid list Y_k. To diversify the recommendations, we compute pairwise Euclidean distances between cluster centroids and identify the furthest sets $\mathcal{Q}_{\text{far}}$. We freeze the top-K_f highest-affinity recommendations to maintain user personalization. For the cluster centroid, tail items ($> K_f$) that overlap with recommendations from distant centroids $\mathcal{Q}_{\text{far}}$ are prioritized by cluster distance, producing a diversified centroid list $\hat{Y}_k$. For an individual user, we keep their top-K_f recommendations fixed to preserve personalization, while their remaining tail items are filtered to retain only those present in $\hat{Y}_k$. Computing this reordering once per cluster centroid reduces computational complexity from $O(N)$ across all users to $O(K_c)$ across clusters.

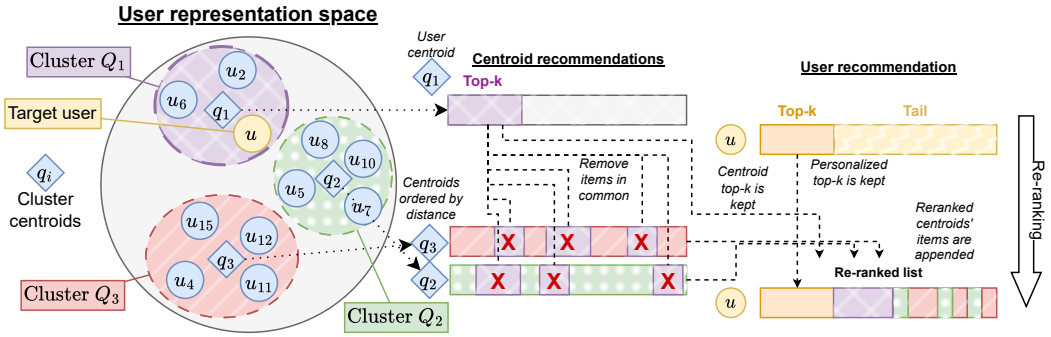


Figure 2: Each cluster is represented by the user nearest to its centroid. A target user's top-K recommendations are frozen to preserve relevance, while tail items are reordered based on their overlap with distant cluster centroids and sorted by score.

3. Experimental Setup

Dataset. We validate our framework on an industrial dataset from TIM S.p.A [9]. Interactions follow a severe long-tail distribution: 83.05% of users have never made a purchase (extremely cold), 15.68% have interacted with only one item (cold), and 1.26% represent active customers (warm). Correspondingly, 36.36% of catalog services have zero sales history.

Evaluation Metrics. To assess the balance between personalization, catalog exploration, and recommendation diversity, we employ three metrics: Intra-List Diversity (ILD@K), Coverage@K, and Type-Coverage@K. The first one computes mean pairwise Euclidean distance between latent embeddings of recommended items, the Coverage@K indicates the fraction of unique items appearing across all top-K recommendation lists (Cov@K $= |\bigcup_u Y_K(u)|/|I|$), and the last one evaluates coverage stratified across catalog tiers: Warm (W@K), Cold (C@K), and Unsold (U@K) items.

Baselines. We compare our method against three baselines: (i) Base GNN, which relies on direct dot-product scoring without diversity re-ranking; (ii) Random-Shifted GNN (RS-GNN), introducing stochastic permutations to candidate lists; and (iii) Popularity-Shifted GNN (PS-GNN), which sorts items by interaction frequency.

Configuration. Hyperparameters are selected via Bayesian optimization: PCA components $m \in \{100, 150, 300\}$, selected SHAP features $d \in \{5, 10, 20\}$, Annoy neighbors $k \in \{3, 10, 20\}$, and frozen depth $K_f \in \{5, 10, 20, 50\}$. Final configurations are denoted as (m, d, k).

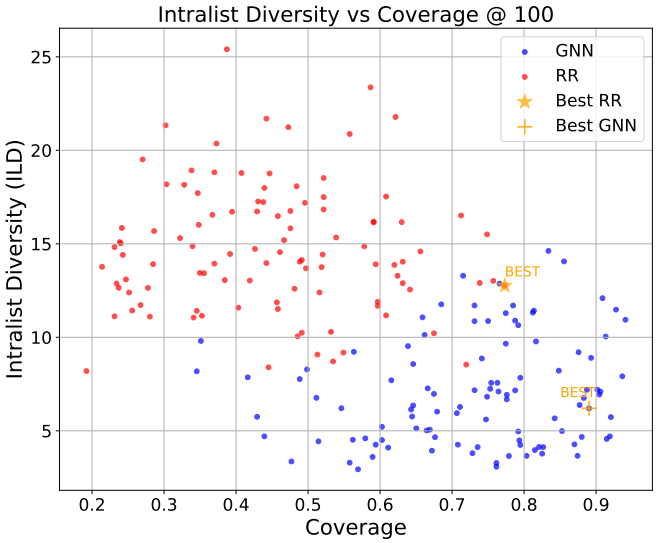


Figure 3: Catalog Coverage vs. Intra-List Diversity (ILD) at $K = 100$ across all explored hyperparameter configurations. Raw GNN outputs (blue) and re-ranked models (red) form two clearly separated clusters. The selected production setting ('BEST', yellow marker) maximizes catalog exposure along the empirical Pareto frontier.

4. Results

To evaluate the dynamics between catalog exposure and recommendation diversity, Fig. 3 maps Coverage@100 against Intra-List Diversity (ILD@100) across all explored hyperparameter configurations. The empirical outcomes visibly are separated into two distinct clusters. Raw GNN predictions (blue markers) occupy the bottom-right space, achieving broad catalog coverage exceeding 90% but suffering from low intra-list diversity (ILD < 15). While the unconstrained GNN exposes a wide variety of items across the global customer base, recommendations presented to any single user remain largely homogeneous. Conversely, applying our re-ranking strategy (red markers) shifts ILD values up to 25.4. This diversification incurs an expected trade-off: steering recommendations toward distant latent regions naturally consolidates the global candidate pool, slightly lowering aggregate coverage across constrained list lengths.

Crucially, the spread of configurations in Fig. 3 demonstrates the flexibility of our architecture in industrial practice, mapping out an adjustable Pareto frontier rather than a single rigid outcome. The curve can be navigated according to specific industrial priorities. For example, prioritizing extreme novelty allows the system to reach higher ILD, though overall catalog coverage below 40%.

The main operational goal is promoting unsold items while maintaining popular item visibility. Guided by this priority, we select the re-ranked recommendations, designated as **'BEST'** in Fig. 3. This configuration uses 100 PCA dimensions, 20 SHAP attributes, and 3 augmentation neighbors under RR-Frozen@50.

5. Conclusion

In this paper, we addressed the extreme cold-start challenge in industrial recommender systems by introducing an end-to-end framework combining XAI-guided graph densification, inductive GNN representation learning, and a diversity re-ranking algorithm. Validated on a production telecommunications catalog covering over one million users, our framework enhances Intra-List Diversity while maintaining an optimal balance of catalog coverage along the empirical Pareto frontier. This unified framework offers a scalable, controllable method for deploying diverse graph-based recommendations in real-world environments.

Declaration on Generative AI

During the preparation of this work, the authors used generative AI tools in order to improve writing style and Grammar and spelling check. After using these tools, the authors reviewed and edited the content as needed and takes full responsibility for the publication's content.

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